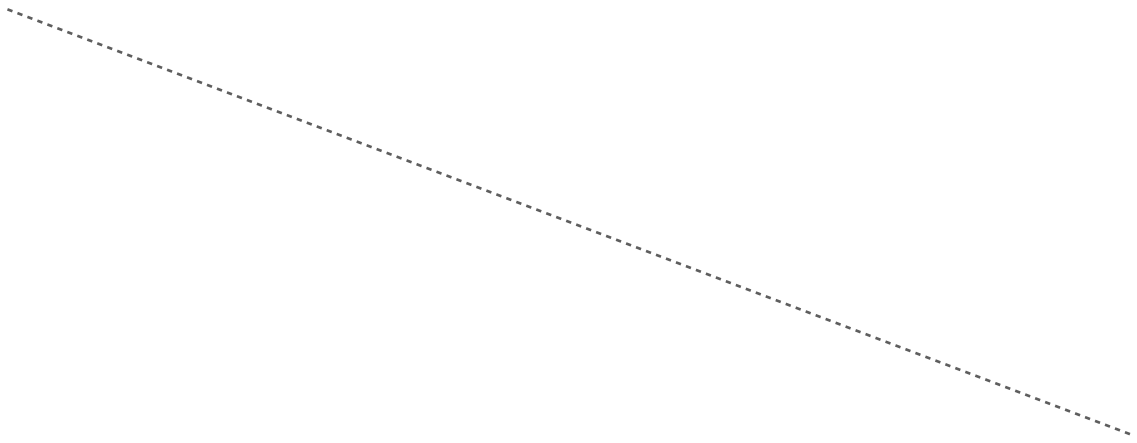
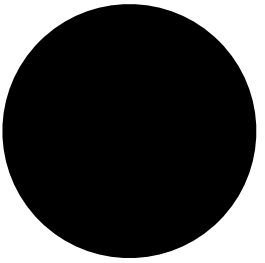
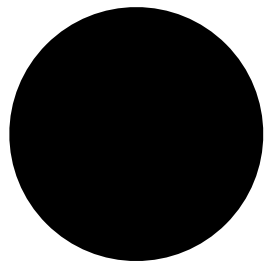
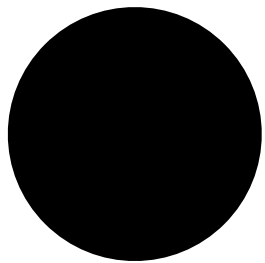
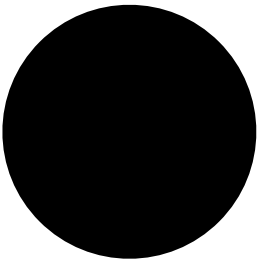
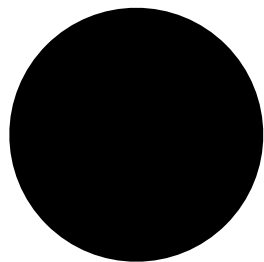
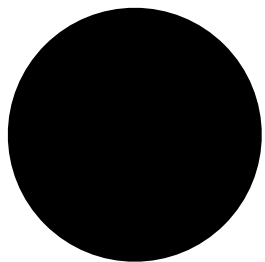
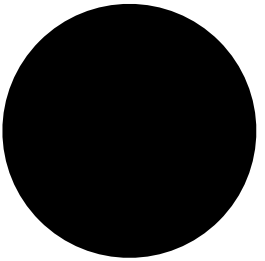
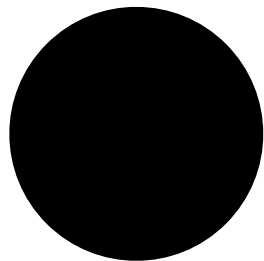
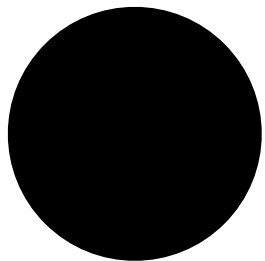
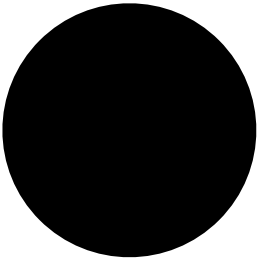
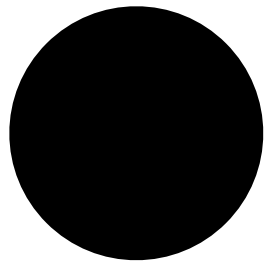
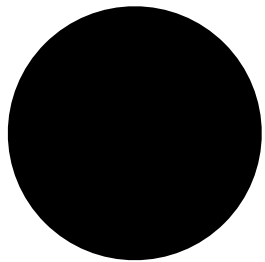
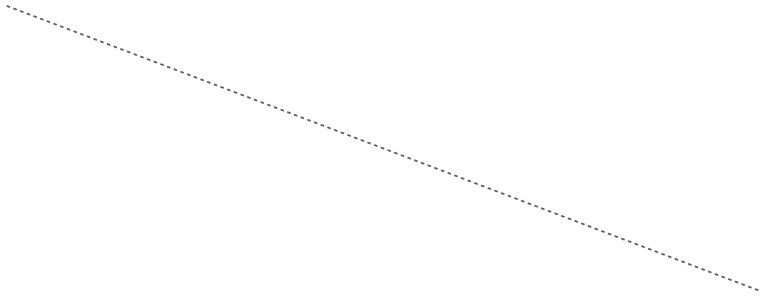
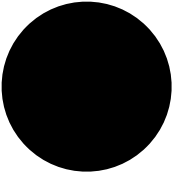
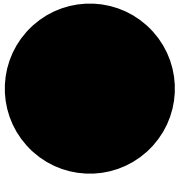
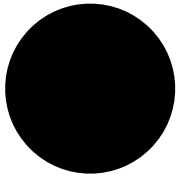
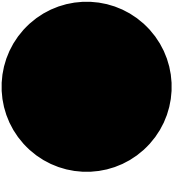
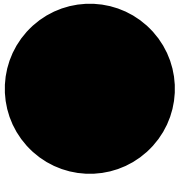
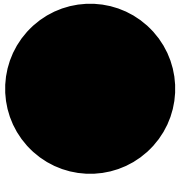
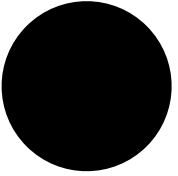
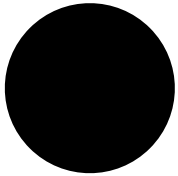
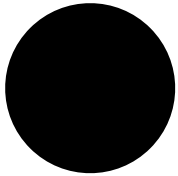
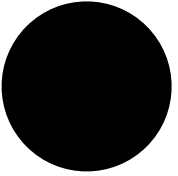
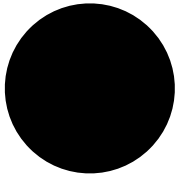
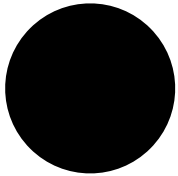
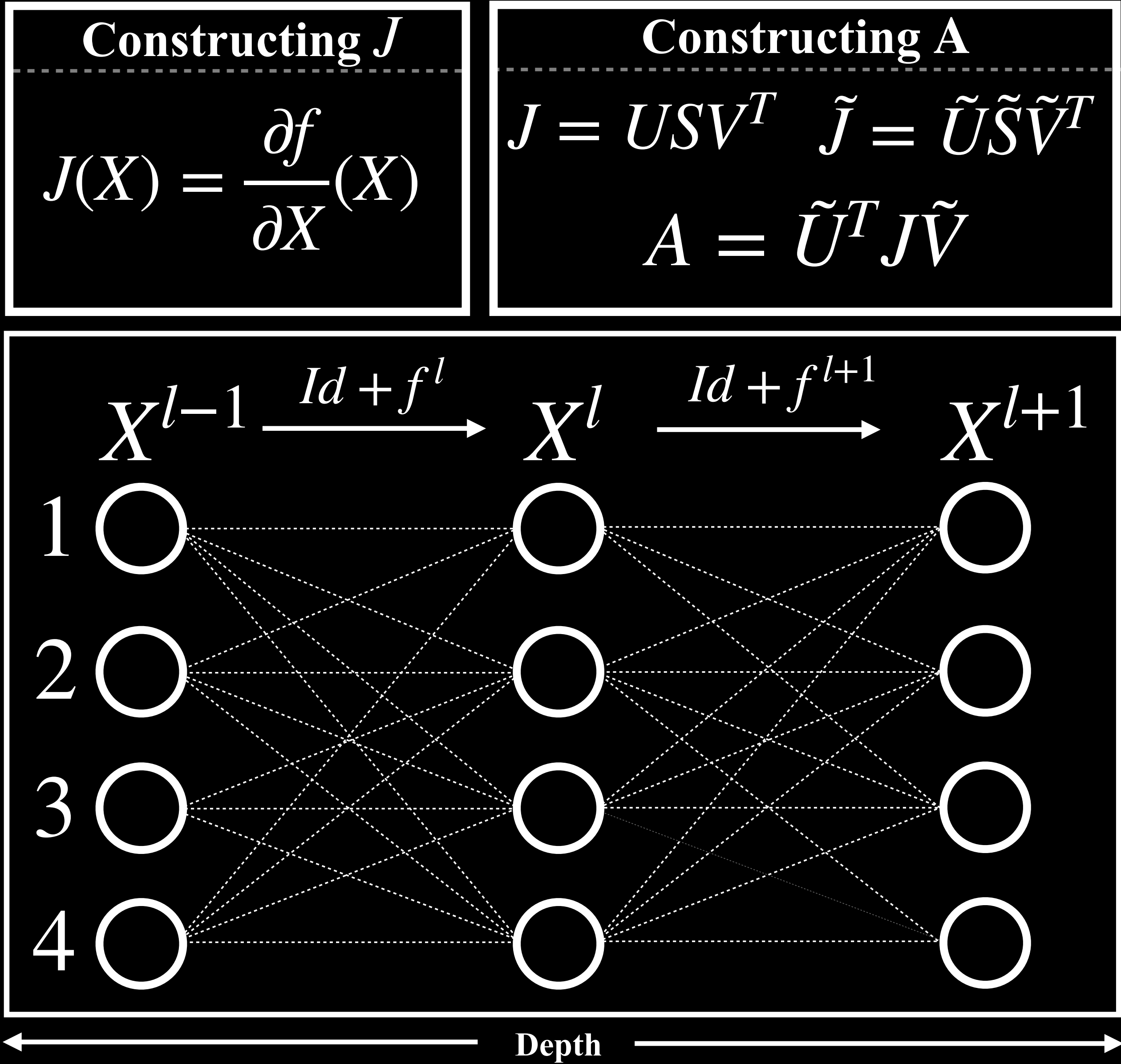








Transformer Block Coupling



$\tilde{U} \overset{?}{\sim} U$

$\tilde{V} \overset{?}{\sim} V$

$\downarrow$

$\tilde{U}^T U \overset{?}{=} I$

$\tilde{V}^T V \overset{?}{=} I$

$A = \tilde{U}^T J \tilde{V}$

$= \tilde{U}^T U S V^T \tilde{V}$

$\overset{?}{=} S$

$m(A) = \frac{\|A - S\|_F}{\|A\|_F}$

Constructing J

Constructing A

$$J(X) = \frac{\partial f}{\partial X}(X)$$

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

Depth-wise Coupling

$$J_{11}^l = \frac{\partial}{\partial x_1^{l-1}} \left(f^l(X^{l-1}) \right)_1$$

$$J_{11}^{l+1} = \frac{\partial}{\partial x_1^l} \left(f^{l+1}(X^l) \right)_1$$

$\left. \vphantom{\frac{\partial}{\partial x_1^{l-1}}} \right\} A_{11}^{l,l+1}$

Tokens

Depth